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CHALLENGE 03

Housing Typology, Equity & Climate Matchmaker

Build a decision-support tool that matches places with plausible housing types and explains the tradeoffs behind each scenario.

HOW TO USE THESE BRIEFS

These briefs consolidate the original challenge set and the additional ideas developed in review. Each is broad enough to support multiple approaches while still naming a concrete user, decision, and demo.

Teams should build a working prototype, use real regional data where possible, explain assumptions and uncertainty, and demonstrate the product with at least one



Pittsburgh or Allegheny County use case.

CORE PROBLEM

Communities need more homes, but the best form and location vary with household demand, land, infrastructure, transit, affordability, and climate impacts. A simple label such as "missing middle" does not tell a planner, nonprofit, or developer whether a particular neighborhood needs duplexes, apartments, townhomes, accessory units, senior housing, or another option, or whether current zoning allows it.

BUILD CHALLENGE

Build a decision-support tool that matches places with plausible housing types and explains the tradeoffs. The model should compare demand, physical feasibility, affordability, displacement risk, infrastructure capacity, access to opportunity, and marginal carbon emissions. It should present scenarios rather than declare a single objectively correct neighborhood or housing type.

SUCCESS LOOKS LIKE

A user can compare at least two housing scenarios for a real place, see why the tool ranked them differently, change normative weights, and understand which conclusions are data-driven versus value judgments.

Owner / input

Housing-policy and planning leads with community-stakeholder input.

Primary users

- Municipal planners testing zoning and infrastructure scenarios
- Community development corporations choosing projects that meet local needs
- Developers evaluating product type and likely market demand
- Residents and public officials comparing alternative growth patterns

Prototype possibilities

- Neighborhood-to-typology match with interpretable factors and confidence ranges
- Scenario tool comparing duplexes, townhomes, apartments, accessory units, and detached homes
- Climate score based on building form, embodied carbon, transportation, and infrastructure extension
- Equity dashboard identifying affordability gaps and access to jobs, schools, transit, and services
- Policy simulator for zoning, tax incentives, density bonuses, or infrastructure investments

Useful data

- HUD CHAS
- Land-use data
- Transit accessibility
- Demographic indicators

- Environmental resilience layers

READY TO BUILD?

Use this brief to compare housing scenarios for a real place.

The strongest submissions will make tradeoffs legible and separate observed evidence from policy choices, assumptions, and value judgments.

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